Plate and Shell Elements

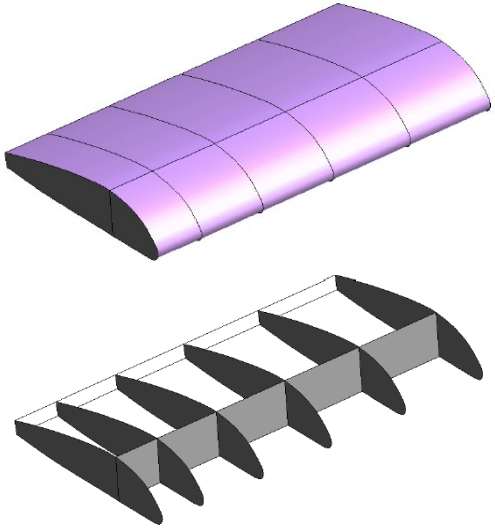
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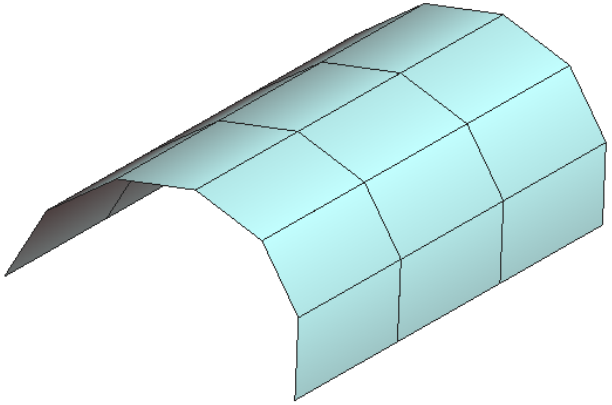
Motivation



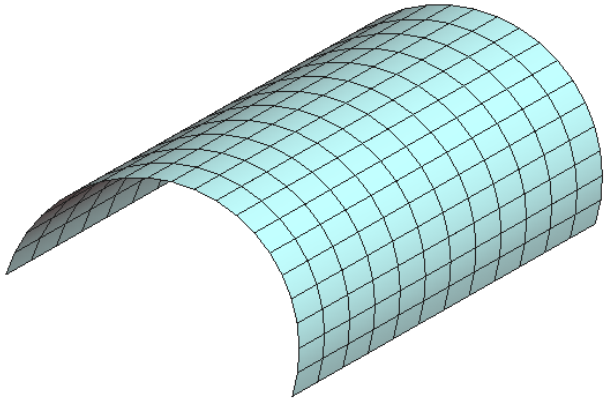
Fuselage



Structural elements of the wing



Coarse mesh

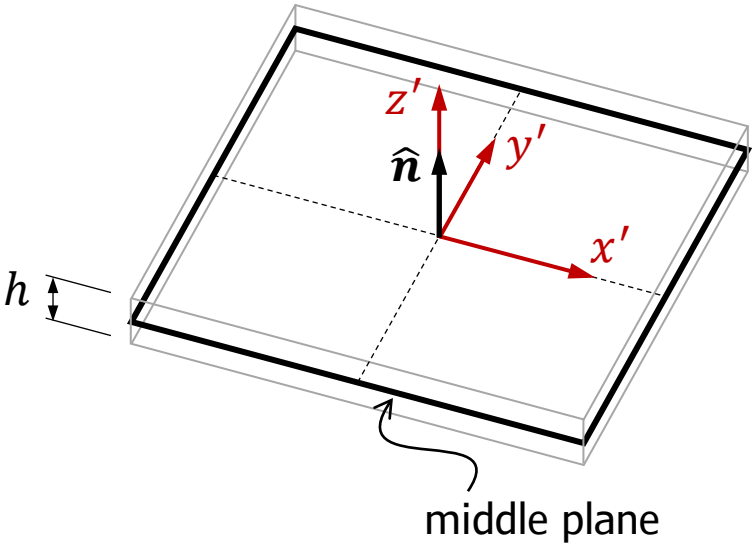


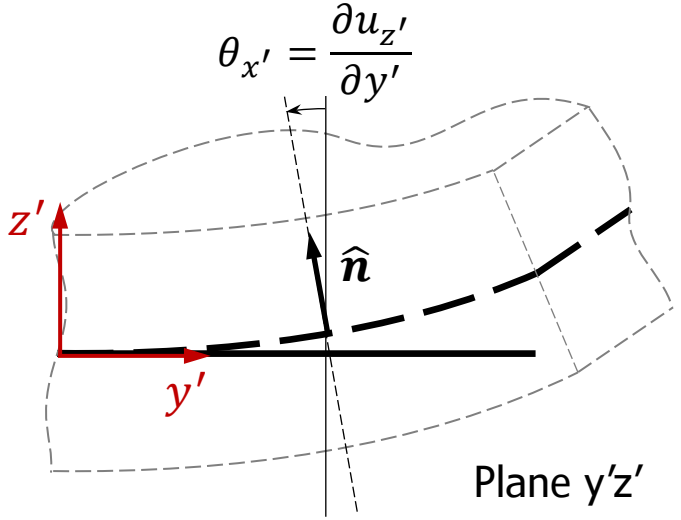
Fine mesh

Flat shell theory

Hypothesis for *Kirchhoff thin plate theory*:

1) Points along the normal to the middle plane have the same vertical displacement (the thickness remains constant during deformation).
2) Normal stress component is negligible, $\sigma_{z'} \approx 0$ (***plane stress*** assumption).
3) A straight line normal to the undeformed middle plane remains *straight* after deformation.
4) Points on the middle plane only move vertically ($\bar{u}_{x'} = \bar{u}_{y'} = 0$).
5) A straight line normal to the undeformed middle plane remains *normal* to the deformed middle plane.



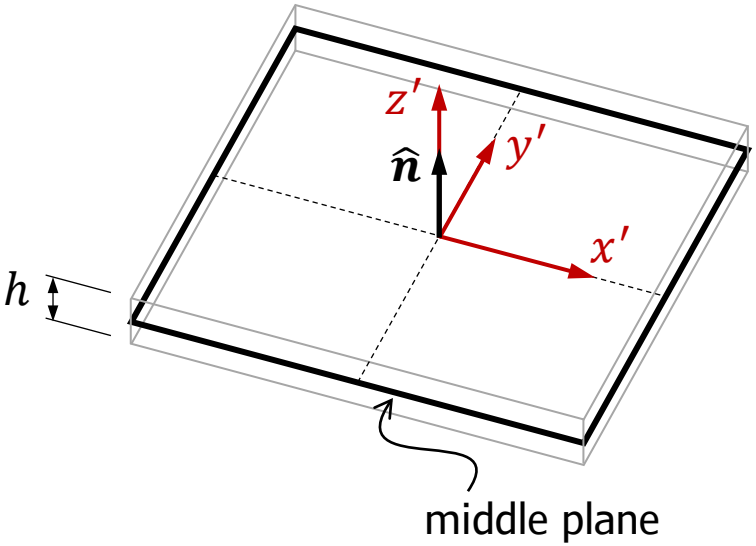


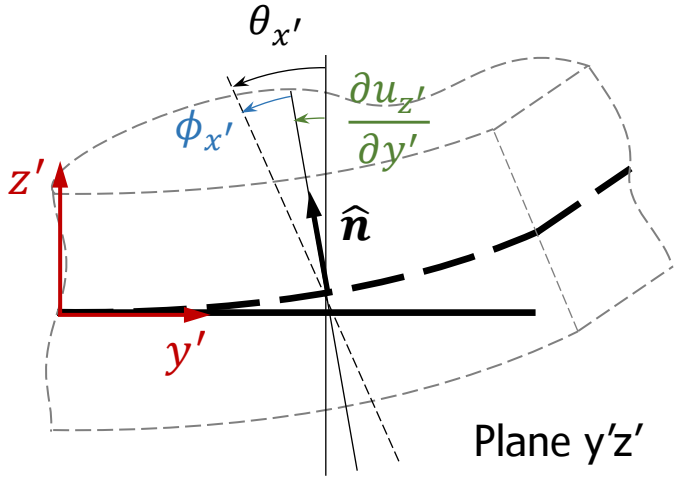
Flat shell theory

Hypothesis for *Reissner-Mindlin plate theory*:

1) Points along the normal to the middle plane have the same vertical displacement (the thickness remains constant during deformation).
2) Normal stress component is negligible, $\sigma_{z'} \approx 0$ (***plane stress*** assumption).
3) A straight line normal to the undeformed middle plane remains *straight* after deformation.
4) Points on the middle plane only move vertically ($\bar{u}_{x'} = \bar{u}_{y'} = 0$).

~~5) A straight line normal to the undeformed middle plane remains *normal* to the deformed middle plane.~~





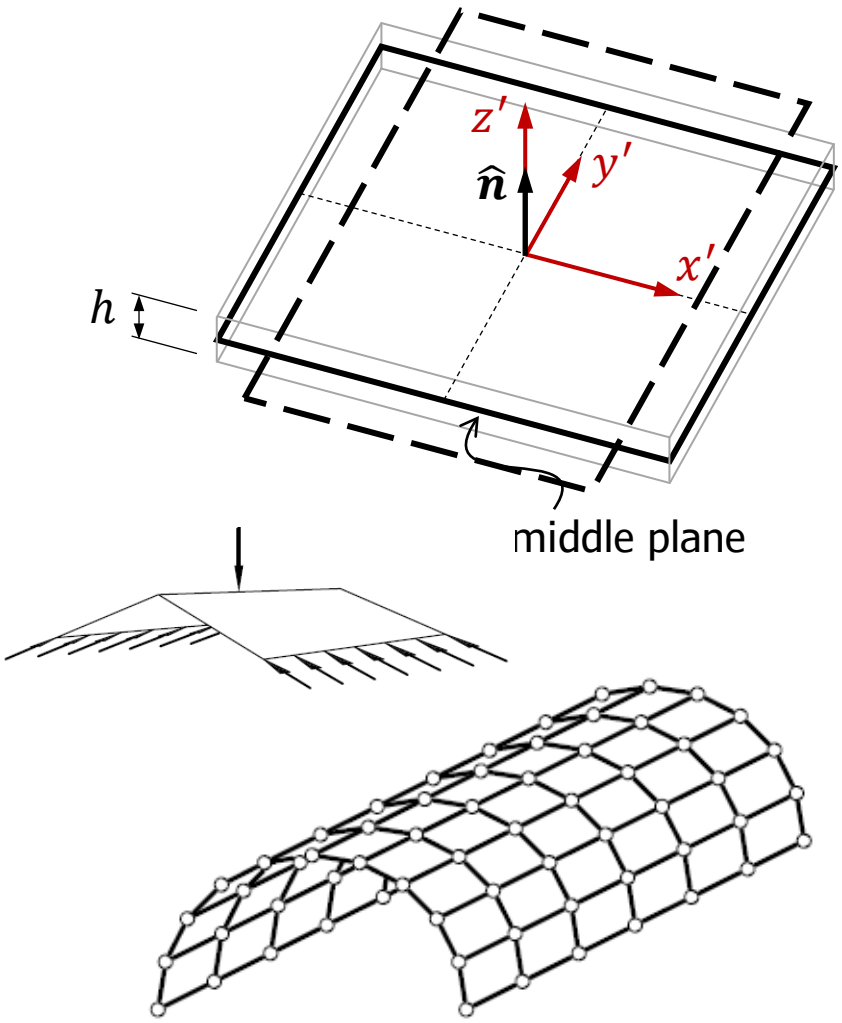
Flat shell theory

Hypothesis for *Flat shell theory*:

1) Points along the normal to the middle plane have the same vertical displacement (the thickness remains constant during deformation).
2) Normal stress component is negligible, $\sigma_{z'} \approx 0$ (***plane stress*** assumption).
3) A straight line normal to the undeformed middle plane remains *straight* after deformation.

~~4) Points on the middle plane only move vertically ($\bar{u}_{x'} = \bar{u}_{y'} = 0$).~~

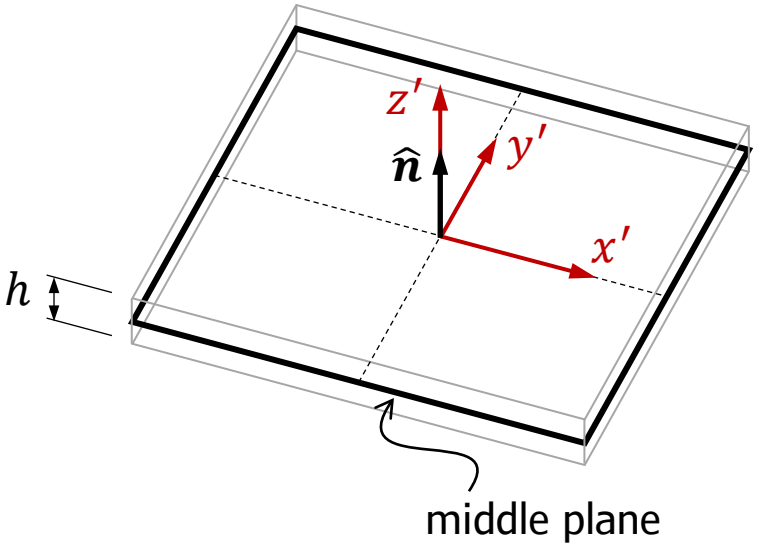
~~5) A straight line normal to the undeformed middle plane remains *normal* to the deformed middle plane.~~



Flat shell theory

Local displacements:

$$u_{x'}(\boldsymbol{x}') = \bar{u}_{x'}(x', y') + z'\theta_{y'}(x', y')$$
$$u_{y'}(\boldsymbol{x}') = \bar{u}_{y'}(x', y') - z'\theta_{x'}(x', y')$$
$$u_{z'}(\boldsymbol{x}') = \bar{u}_{z'}(x', y')$$



Local strain:

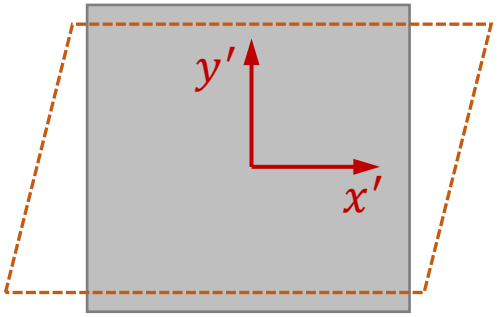
$$\varepsilon_{x'x'} = \frac{\partial u_{x'}}{\partial x'} = \frac{\partial \bar{u}_{x'}}{\partial x'} + z'\frac{\partial \theta_{y'}}{\partial x'}$$
$$\varepsilon_{y'y'} = \frac{\partial u_{y'}}{\partial y'} = \frac{\partial \bar{u}_{y'}}{\partial y'} - z'\frac{\partial \theta_{x'}}{\partial y'}$$
$$\varepsilon_{z'z'} = \frac{\partial u_{z'}}{\partial z'} = 0 \longrightarrow \text{Hypothesis (1)}$$
$$\gamma_{x'y'} = \frac{\partial u_{y'}}{\partial x'} + \frac{\partial u_{x'}}{\partial y'} = \frac{\partial \bar{u}_{y'}}{\partial x'} + \frac{\partial \bar{u}_{x'}}{\partial y'} + z'\left(\frac{\partial \theta_{y'}}{\partial y'} - \frac{\partial \theta_{x'}}{\partial x'}\right)$$
$$\gamma_{x'z'} = \frac{\partial u_{z'}}{\partial x'} + \frac{\partial u_{x'}}{\partial z'} = \frac{\partial \bar{u}_{z'}}{\partial x'} + \theta_{y'} = -\phi_{y'}$$
$$\gamma_{y'z'} = \frac{\partial u_{z'}}{\partial y'} + \frac{\partial u_{y'}}{\partial z'} = \frac{\partial \bar{u}_{z'}}{\partial y'} - \theta_{x'} = -\phi_{x'}$$

- If hypothesis (4) holds, red terms vanish.
- If hypothesis (5) holds, $\gamma_{x'z'} = \gamma_{y'z'} = 0$.

Flat shell theory

Local displacements:

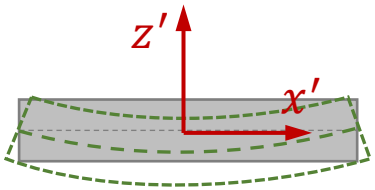
$$u_{x'} = \bar{u}_{x'} + z'\theta_{y'}$$
$$u_{y'} = \bar{u}_{y'} - z'\theta_{x'}$$
$$u_{z'} = \bar{u}_{z'}$$



Membrane (in-plane)

Local strain:

$$\varepsilon_{x'x'} = \frac{\partial u_{x'}}{\partial x'} = \frac{\partial \bar{u}_{x'}}{\partial x'} + z'\frac{\partial \theta_{y'}}{\partial x'}$$
$$\varepsilon_{y'y'} = \frac{\partial u_{y'}}{\partial y'} = \frac{\partial \bar{u}_{y'}}{\partial y'} - z'\frac{\partial \theta_{x'}}{\partial y'}$$



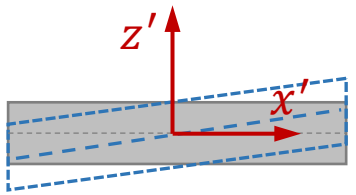
Bending

$$\varepsilon_{z'z'} = \frac{\partial u_{z'}}{\partial z'} = 0$$

$$\gamma_{x'y'} = \frac{\partial u_{y'}}{\partial x'} + \frac{\partial u_{x'}}{\partial y'} = \frac{\partial \bar{u}_{y'}}{\partial x'} + \frac{\partial \bar{u}_{x'}}{\partial y'} + z'\left(\frac{\partial \theta_{y'}}{\partial y'} - \frac{\partial \theta_{x'}}{\partial x'}\right)$$

$$\gamma_{x'z'} = \frac{\partial u_{z'}}{\partial x'} + \frac{\partial u_{x'}}{\partial z'} = \frac{\partial \bar{u}_{z'}}{\partial x'} + \theta_{y'} = -\phi_{y'}$$

$$\gamma_{y'z'} = \frac{\partial u_{z'}}{\partial y'} + \frac{\partial u_{y'}}{\partial z'} = \frac{\partial \bar{u}_{z'}}{\partial y'} - \theta_{x'} = -\phi_{x'}$$



Shear

Flat shell theory

(*) Note on notation: $\dfrac{\partial a_i}{\partial x_j} = a_{i,j}$

Local displacements (vector form):

$$\{\boldsymbol{u}'\} = \begin{Bmatrix} u_{x'} \\ u_{y'} \\ u_{z'} \end{Bmatrix} = \overbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & z' \\ 0 & 1 & 0 & -z' & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}}^{[\mathbf{D}(z')]} \overbrace{\begin{Bmatrix} \bar{u}_{x'} \\ \bar{u}_{y'} \\ \bar{u}_{z'} \\ \theta_{x'} \\ \theta_{y'} \end{Bmatrix}}^{\{\bar{\boldsymbol{u}}'(x',y')\}}$$

Local strain (vector form) decomposition:

$$\{\boldsymbol{\varepsilon}'\} = \begin{Bmatrix} \varepsilon_{x'x'} \\ \varepsilon_{y'y'} \\ \gamma_{x'y'} \\ \gamma_{x'z'} \\ \gamma_{y'z'} \end{Bmatrix} = \overbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & z' & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -z' & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & z' \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}}^{[\mathbf{S}(z')]} \overbrace{\begin{Bmatrix} \bar{u}_{x',x'} \\ \bar{u}_{y',y'} \\ \bar{u}_{x',y'} + \bar{u}_{y',x'} \\ \bar{u}_{z',x'} + \theta_{y'} \\ \bar{u}_{z',y'} - \theta_{z'} \\ \theta_{y',x'} \\ \theta_{x',y'} \\ \theta_{y',y'} - \theta_{x',x'} \end{Bmatrix}}^{\{\bar{\boldsymbol{\varepsilon}}'(x',y')\}}$$

Flat shell theory

Local stresses (vector form):

$$\{\boldsymbol{\sigma}'\} = \begin{Bmatrix} \sigma_{x'x'} \\ \sigma_{y'y'} \\ \tau_{x'y'} \\ \tau_{x'z'} \\ \tau_{y'z'} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & 0 & 0 & 0 \\ C_{21} & C_{22} & 0 & 0 & 0 \\ 0 & 0 & C_{33} & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & C_{55} \end{bmatrix} \begin{Bmatrix} \varepsilon_{x'x'} \\ \varepsilon_{y'y'} \\ \gamma_{x'y'} \\ \gamma_{x'z'} \\ \gamma_{y'z'} \end{Bmatrix} = [\mathbf{C}']\{\boldsymbol{\varepsilon}'\}$$

$[\mathbf{C}'_{\mathrm{p}}]$: Plane-stress constitutive relation

$[\mathbf{C}'_{\mathrm{s}}]$: Shear stress-strain constitutive relation

Constitutive matrices:

$$[\mathbf{C}'_{\mathrm{p}}] = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}; \qquad [\mathbf{C}'_{\mathrm{s}}] = k_{\mathrm{s}} G \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad k_{\mathrm{s}} = \frac{5}{6}$$

with E being the Young's modulus, ν the Poisson's ratio and G the shear modulus. For linear isotropic materials: $G = E/\big(2(1+\nu)\big)$

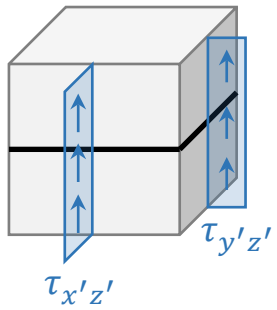
Flat shell theory

Local stresses (vector form):

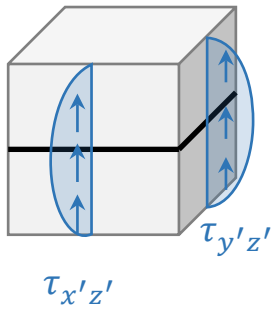
$$\{\boldsymbol{\sigma}'\} = \begin{Bmatrix} \sigma_{x'x'} \\ \sigma_{y'y'} \\ \tau_{x'y'} \\ \tau_{x'z'} \\ \tau_{y'z'} \end{Bmatrix} = \begin{bmatrix} C_{11} \\ C_{21} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Plane-stress constitutive relation

Shear stresses according to Reissner-Mindlin theory



Exact shear stresses distribution



Constitutive matrices:

$$[\mathbf{C}'_{\mathrm{p}}] = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}; \qquad [\mathbf{C}'_{\mathrm{s}}] = k_{\mathrm{s}} G \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \qquad k_{\mathrm{s}} = \frac{5}{6}$$

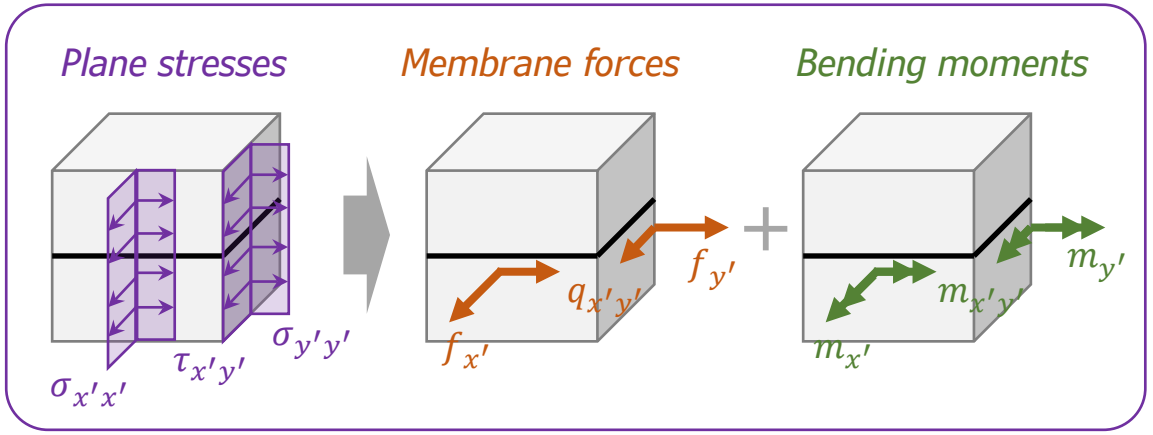
Shear correction factor

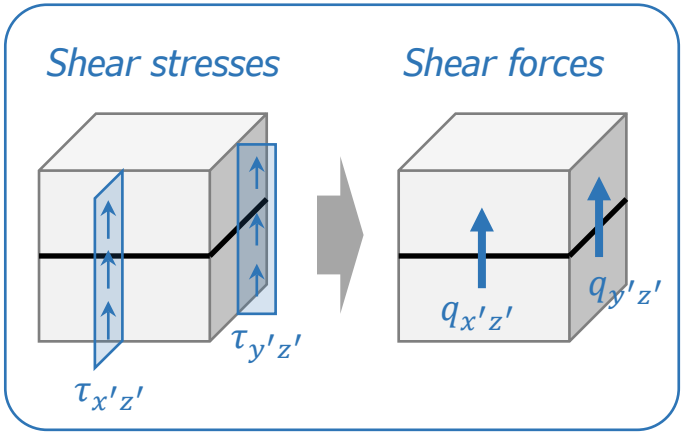
with E being the Young's modulus, ν the Poisson's ratio and G the shear modulus. For linear isotropic materials: $G = E/\big(2(1+\nu)\big)$

Flat shell theory

Decomposed internal forces between plates:

$$\{\bar{f}'_{\text{int}}\} = \begin{Bmatrix} f_{x'} \\ f_{y'} \\ q_{x'y'} \\ q_{x'z'} \\ q_{y'z'} \\ m_{y'} \\ m_{x'} \\ m_{x'y'} \end{Bmatrix} = \int_{-h/2}^{h/2} \begin{Bmatrix} \sigma_{x'x'} \\ \sigma_{y'y'} \\ \tau_{x'y'} \\ \tau_{x'z'} \\ \tau_{x'y'} \\ z'\sigma_{x'x'} \\ -z'\sigma_{y'y'} \\ z'\tau_{x'y'} \end{Bmatrix} dz' = \int_{-h/2}^{h/2} \overbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ z' & 0 & 0 & 0 & 0 \\ 0 & -z' & 0 & 0 & 0 \\ 0 & 0 & z' & 0 & 0 \end{bmatrix}}^{[\mathbf{S}(z')]^{\mathrm{T}}} \begin{Bmatrix} \sigma_{x'x'} \\ \sigma_{y'y'} \\ \tau_{x'y'} \\ \tau_{x'z'} \\ \tau_{y'z'} \end{Bmatrix} \mathrm{d}z'$$





Flat shell theory

Equilibrium equation (strong form):

$$\dot{\boldsymbol{p}}' = \boldsymbol{\nabla}_{\boldsymbol{x}'} \cdot \boldsymbol{\sigma}' + \rho \boldsymbol{b}' + \boldsymbol{q}'$$

where $\boldsymbol{p}' = \rho \dot{\boldsymbol{u}}'$ is the linear momentum, ρ is the density (kg/m^3), $\boldsymbol{b}'$ are external body forces (N/kg) and $\boldsymbol{q}'$ are volume forces (N/m^3). The set of boundary conditions of the problem are:

- Prescribed displacement: $\boldsymbol{u}'|_{\boldsymbol{x}' \in \Gamma_u} = \boldsymbol{u}'_{\mathrm{p}}$
- External traction force: $\boldsymbol{\sigma}' \cdot \hat{\boldsymbol{n}}|_{\boldsymbol{x}' \in \Gamma_\sigma} = \boldsymbol{t}'$

Principle of virtual work (weak form):

$$\int_\Omega \{\delta \boldsymbol{\varepsilon}'\}^{\mathrm{T}} \{\boldsymbol{\sigma}'\} \mathrm{d}\Omega + \int_\Omega \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \rho \{\ddot{\boldsymbol{u}}'\} \mathrm{d}\Omega = \int_\Omega \{\delta \boldsymbol{u}'\}^{\mathrm{T}} (\rho \{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) \mathrm{d}\Omega + \int_{\Gamma_\sigma} \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \{\boldsymbol{t}'\} \mathrm{d}\Gamma$$

For all $\delta \boldsymbol{u}'|_{\boldsymbol{x}' \in \Gamma_u} = \boldsymbol{0}$, and such that $\{\delta \boldsymbol{\varepsilon}'\}^{\mathrm{T}} = \left\{\delta \varepsilon_{x',x'},\ \delta \varepsilon_{y',y'},\ \delta \gamma_{x',y'},\ \delta \gamma_{x',z'}, \delta \gamma_{y',z'}\right\}$.

Assuming:

$$\{\delta \boldsymbol{u}'\} = [\mathbf{D}(z')]\{\delta \overline{\boldsymbol{u}}'(x', y')\} \quad \text{and} \quad \{\delta \boldsymbol{\varepsilon}'\} = [\mathbf{S}(z')]\{\delta \overline{\boldsymbol{\varepsilon}}'(x', y')\}$$

Flat shell theory

Focusing on each integral on the left-hand side:

- Stiffness term:

$$\int_{\Omega} \{\delta\boldsymbol{\varepsilon}'\}^{\mathrm{T}}\{\boldsymbol{\sigma}'\}\mathrm{d}\Omega = \int_{S} \{\delta\bar{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}} \left(\int_{-h/2}^{h/2} [\mathbf{S}]^{\mathrm{T}}[\mathbf{C}'][\mathbf{S}] dz' \right) \{\bar{\boldsymbol{\varepsilon}}'\}\mathrm{d}S$$

For isotropic materials (constant $[\mathbf{C}']$ over z'):

$$[\bar{\mathbf{C}}'] = \int_{-h/2}^{h/2} [\mathbf{S}]^{\mathrm{T}}[\mathbf{C}'][\mathbf{S}]\mathrm{d}z' = h \begin{bmatrix} C_{11} & C_{12} & 0 & 0 & 0 & z_0' C_{11} & -z_0' C_{12} & 0 \\ C_{21} & C_{22} & 0 & 0 & 0 & z_0' C_{21} & -z_0' C_{22} & 0 \\ 0 & 0 & C_{33} & 0 & 0 & 0 & 0 & z_0' C_{33} \\ 0 & 0 & 0 & k_s C_{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & k_s C_{55} & 0 & 0 & 0 \\ z_0' C_{11} & z_0' C_{12} & 0 & 0 & 0 & r_h'^2 C_{11} & r_h'^2 C_{12} & 0 \\ -z_0' C_{21} & -z_0' C_{22} & 0 & 0 & 0 & r_h'^2 C_{21} & r_h'^2 C_{22} & 0 \\ 0 & 0 & z_0' C_{33} & 0 & 0 & 0 & 0 & r_h'^2 C_{33} \end{bmatrix}$$

$$C_{11} = C_{22} = \frac{E}{1-\nu^2}; \quad C_{12} = C_{21} = \frac{E\nu}{1-\nu^2}; \quad C_{33} = C_{44} = C_{55} = \frac{E}{2(1+\nu)} = G \ \ (= C_{\mathrm{t}}^*)$$

$$r_h'^2 = \frac{1}{h}\int_{-h/2}^{h/2} z'^2 \mathrm{d}z' = \frac{h^2}{12}; \quad z_0' = \frac{1}{h}\int_{-h/2}^{h/2} z' \mathrm{d}z' = 0$$

Bending and membrane stresses and strains are uncoupled!!

Flat shell theory

- Inertial term:

$$\int_{\Omega} \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \rho \{\ddot{\boldsymbol{u}}'\} \mathrm{d}\Omega = \int_{S} \{\delta \overline{\boldsymbol{u}}'\}^{\mathrm{T}} \left(\int_{-h/2}^{h/2} [\mathbf{D}]^{\mathrm{T}} \rho [\mathbf{D}] dz' \right) \{\ddot{\overline{\boldsymbol{u}}}'\} \mathrm{d}S$$

For <u>isotropic materials</u> (constant density ρ over z'):

$$[\overline{\boldsymbol{\rho}}'] = \int_{-h/2}^{h/2} [\mathbf{D}]^{\mathrm{T}} \rho [\mathbf{D}] \mathrm{d}z' = \rho h \begin{bmatrix} 1 & 0 & 0 & 0 & z_0' \\ 0 & 1 & 0 & -z_0' & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -z_0' & 0 & {r_h'}^2 & 0 \\ z_0' & 0 & 0 & 0 & {r_h'}^2 \end{bmatrix}$$

<u>Remark</u>: If the centre of mass is located at the origin of the local reference frame ($z' = 0$), then $z_0 = 0$.

Flat shell theory

Focusing on the right hand side of the equation:

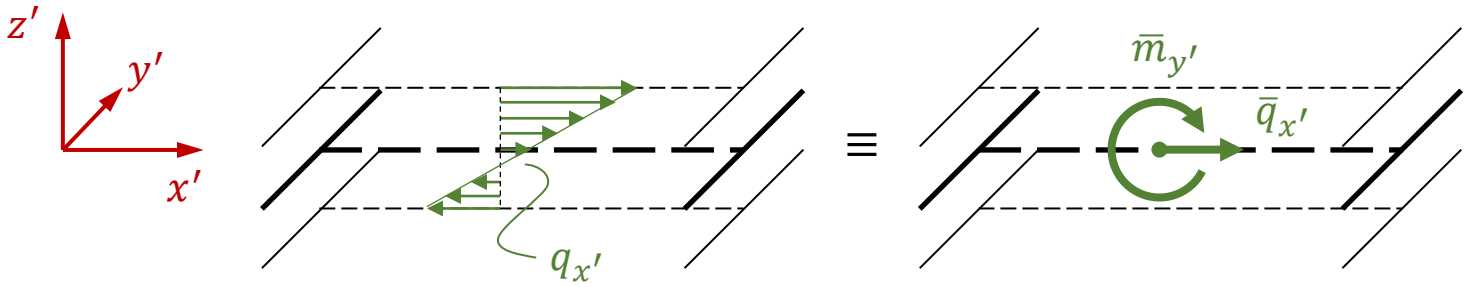
- External body/volume forces (assuming $\boldsymbol{b}'$ is uniform over the thickness):

$$\int_{\Omega} \{\delta \boldsymbol{u}'\}^{\mathrm{T}} (\rho\{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) \mathrm{d}\Omega = \int_{S} \{\delta \overline{\boldsymbol{u}}'\}^{\mathrm{T}} \left(\int_{-h/2}^{h/2} [\mathbf{D}]^{\mathrm{T}} (\rho\{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) dz' \right) \mathrm{d}S$$

$$\{\overline{\boldsymbol{f}}'\} = \int_{-h/2}^{h/2} [\mathbf{D}]^{\mathrm{T}} (\rho\{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) \mathrm{d}z' = [\overline{\boldsymbol{\rho}}'] \begin{Bmatrix} \bar{b}_{x'} \\ \bar{b}_{y'} \\ \bar{b}_{z'} \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} \bar{q}_{x'} \\ \bar{q}_{y'} \\ \bar{q}_{z'} \\ \bar{m}_{x'} \\ \bar{m}_{y'} \end{Bmatrix} = [\overline{\boldsymbol{\rho}}']\{\overline{\boldsymbol{b}}'\} + \{\overline{\boldsymbol{q}}'\}$$

Distributed loads over shell (N/m^2)

$$\bar{q}_{x'} = \int_{-h/2}^{+h/2} q_{x'} \mathrm{d}z' \,;\; \bar{q}_{y'} = \int_{-h/2}^{+h/2} q_{y'} \mathrm{d}z' \,;\; \bar{q}_{z'} = \int_{-h/2}^{+h/2} q_{z'} \mathrm{d}z' \,;\; \bar{m}_{x'} = \int_{-h/2}^{+h/2} -z' q_{y'} \mathrm{d}z' \,;\; \bar{m}_{y'} = \int_{-h/2}^{+h/2} z' q_{x'} \mathrm{d}z'$$



Flat shell theory

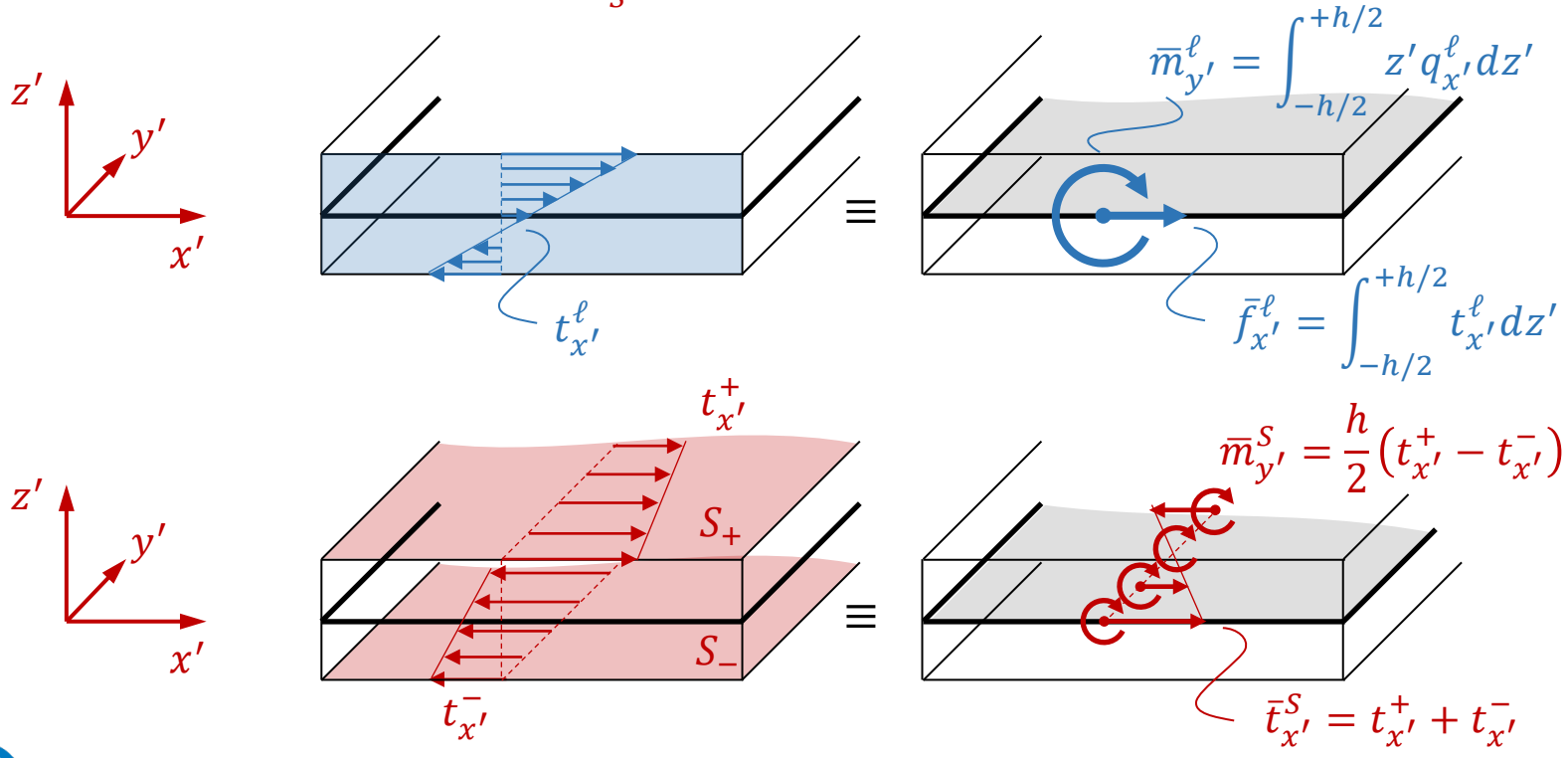
- External surface forces:

Distributed loads over shell (N/m²)

Distributed loads over lines (N/m)

$$\int_{\Gamma_\sigma} \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \{\boldsymbol{t}'\} \mathrm{d}\Gamma = \int_{S^+} \{\delta \boldsymbol{u}'_+\}^{\mathrm{T}} \{\boldsymbol{t}'_+\} \mathrm{d}S + \int_{S^-} \{\delta \boldsymbol{u}'_-\}^{\mathrm{T}} \{\boldsymbol{t}'_-\} \mathrm{d}S + \int_{\ell} \{\delta \bar{\boldsymbol{u}}'_\ell\}^{\mathrm{T}} \left(\int_{-h/2}^{h/2} [\mathbf{D}]^{\mathrm{T}} \{\boldsymbol{t}'_\ell\} \mathrm{d}z' \right) \mathrm{d}\ell$$

$$\int_S \{\delta \bar{\boldsymbol{u}}'\}^{\mathrm{T}} \{\bar{\boldsymbol{t}}'_S\} dS \qquad \{\bar{\boldsymbol{f}}'_\ell\}$$



Flat shell theory

$$\{\bar{\boldsymbol{f}}'_{\mathrm{S}}\} = [\bar{\boldsymbol{\rho}}']\{\bar{\boldsymbol{b}}'\} + \{\bar{\boldsymbol{q}}'\} + \{\bar{\boldsymbol{t}}'_{\mathrm{S}}\}$$

The resulting system yields:

$$\int_S \{\delta\bar{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}}[\bar{\mathbf{C}}']\{\bar{\boldsymbol{\varepsilon}}'\}\mathrm{d}S + \int_S \{\delta\bar{\boldsymbol{u}}'\}^{\mathrm{T}}[\bar{\boldsymbol{\rho}}']\{\ddot{\bar{\boldsymbol{u}}}'\}\mathrm{d}S = \overbrace{\int_S \{\delta\bar{\boldsymbol{u}}'\}^{\mathrm{T}}\{\bar{\boldsymbol{f}}'_{\mathrm{S}}\}\mathrm{d}S}^{\text{Surface loads (N/m}^2\text{)}} + \overbrace{\int_\ell \{\delta\bar{\boldsymbol{u}}'_\ell\}^{\mathrm{T}}\{\bar{\boldsymbol{f}}'_\ell\}\mathrm{d}\ell}^{\text{Line loads (N/m)}} + \overbrace{\{\delta\bar{\boldsymbol{u}}'(x^{(i)})\}^{\mathrm{T}}\{\widehat{\boldsymbol{F}}^{(i)}\}}^{\text{Point loads (N)}}$$

The stiffness term can be further decomposed into different components:

$$\int_S \{\delta\bar{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}}[\bar{\mathbf{C}}']\{\bar{\boldsymbol{\varepsilon}}'\}\mathrm{d}S = \int_S \{\delta\bar{\boldsymbol{\varepsilon}}'_{\mathrm{m}}\}^{\mathrm{T}}[\bar{\mathbf{C}}'_{\mathrm{m}}]\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{m}}\}\mathrm{d}S + \int_S \{\delta\bar{\boldsymbol{\varepsilon}}'_{\mathrm{b}}\}^{\mathrm{T}}[\bar{\mathbf{C}}'_{\mathrm{b}}]\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{b}}\}\mathrm{d}S + \int_S \{\delta\bar{\boldsymbol{\varepsilon}}'_{\mathrm{s}}\}^{\mathrm{T}}[\bar{\mathbf{C}}'_{\mathrm{s}}]\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{s}}\}\mathrm{d}S$$

$$\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{m}}\} = \begin{Bmatrix} \bar{u}_{x',x'} \\ \bar{u}_{y',y'} \\ \bar{u}_{x',y} + \bar{u}_{y',x'} \end{Bmatrix}; \quad [\bar{\mathbf{C}}'_{\mathrm{m}}] = \frac{hE}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}$$

$$\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{b}}\} = \begin{Bmatrix} \theta_{y',x'} \\ \theta_{x',y'} \\ \theta_{y',y'} - \theta_{x',x'} \end{Bmatrix}; \quad [\bar{\mathbf{C}}'_{\mathrm{b}}] = \frac{h^3E}{12(1-\nu^2)}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}$$

$$\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{s}}\} = \begin{Bmatrix} \bar{u}_{z',x'} + \theta_{y'} \\ \bar{u}_{z',y'} - \theta_{x'} \end{Bmatrix}; \quad [\bar{\mathbf{C}}'_{\mathrm{s}}] = \frac{5hE}{12(1+\nu)}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Finite element discretization

Discretized displacements of element (e):

$$\left\{\overline{\boldsymbol{u}}'^{(e)}\right\} = \sum_i \begin{bmatrix} N^{(e,i)} & 0 & 0 & 0 & 0 \\ 0 & N^{(e,i)} & 0 & 0 & 0 \\ 0 & 0 & N^{(e,i)} & 0 & 0 \\ 0 & 0 & 0 & N^{(e,i)} & 0 \\ 0 & 0 & 0 & 0 & N^{(e,i)} \end{bmatrix} \begin{Bmatrix} \hat{u}_{x'}^{(e,i)} \\ \hat{u}_{y'}^{(e,i)} \\ \hat{u}_{z'}^{(e,i)} \\ \hat{\theta}_{x'}^{(e,i)} \\ \hat{\theta}_{y'}^{(e,i)} \end{Bmatrix} = \begin{bmatrix} \dots & \mathbf{N}^{(e,i)} & \dots \end{bmatrix} \begin{Bmatrix} \vdots \\ \widehat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Discretized *membrane* strains of element (e):

$$\left\{\boldsymbol{\varepsilon}'^{(e)}_{\mathrm{m}}\right\} = \sum_i \begin{bmatrix} N^{(e,i)}_{,x'} & 0 & 0 & 0 & 0 \\ 0 & N^{(e,i)}_{,y'} & 0 & 0 & 0 \\ N^{(e,i)}_{,y'} & N^{(e,i)}_{,x'} & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \hat{u}_{x'}^{(e,i)} \\ \hat{u}_{y'}^{(e,i)} \\ \hat{u}_{z'}^{(e,i)} \\ \hat{\theta}_{x'}^{(e,i)} \\ \hat{\theta}_{y'}^{(e,i)} \end{Bmatrix} = \begin{bmatrix} \dots & \mathbf{B}'^{(e,i)}_{\mathrm{m}} & \dots \end{bmatrix} \begin{Bmatrix} \vdots \\ \widehat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Finite element discretization

Discretized *bending* strains of element (e):

$$\left\{\boldsymbol{\varepsilon}'^{(e)}_{\mathrm{b}}\right\} = \sum_i \begin{bmatrix} 0 & 0 & 0 & 0 & N^{(e,i)}_{,x'} \\ 0 & 0 & 0 & N^{(e,i)}_{,y'} & 0 \\ 0 & 0 & 0 & -N^{(e,i)}_{,x'} & N^{(e,i)}_{,y'} \end{bmatrix} \begin{Bmatrix} \hat{u}^{(e,i)}_{x'} \\ \hat{u}^{(e,i)}_{y'} \\ \hat{u}^{(e,i)}_{z'} \\ \hat{\theta}^{(e,i)}_{x'} \\ \hat{\theta}^{(e,i)}_{y'} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{B}'^{(e,i)}_{\mathrm{b}} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Discretized *shear* strains of element (e):

$$\left\{\boldsymbol{\varepsilon}'^{(e)}_{\mathrm{s}}\right\} = \sum_i \begin{bmatrix} 0 & 0 & N^{(e,i)}_{,x'} & 0 & N^{(e,i)} \\ 0 & 0 & N^{(e,i)}_{,y'} & -N^{(e,i)} & 0 \end{bmatrix} \begin{Bmatrix} \hat{u}^{(e,i)}_{x'} \\ \hat{u}^{(e,i)}_{y'} \\ \hat{u}^{(e,i)}_{z'} \\ \hat{\theta}^{(e,i)}_{x'} \\ \hat{\theta}^{(e,i)}_{y'} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{B}'^{(e,i)}_{\mathrm{s}} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Finite element discretization

Discretized *body force/distributed load* vector of element (e):

$$\left\{\bar{\boldsymbol{f}}'^{(e)}_{\mathrm{S}}\right\} = \sum_i \begin{bmatrix} N^{(e,i)} & 0 & 0 & 0 & 0 \\ 0 & N^{(e,i)} & 0 & 0 & 0 \\ 0 & 0 & N^{(e,i)} & 0 & 0 \\ 0 & 0 & 0 & N^{(e,i)} & 0 \\ 0 & 0 & 0 & 0 & N^{(e,i)} \end{bmatrix} \begin{Bmatrix} \hat{f}'^{(e,i)}_{x'} \\ \hat{f}'^{(e,i)}_{y'} \\ \hat{f}'^{(e,i)}_{z'} \\ \hat{m}'^{(e,i)}_{x'} \\ \hat{m}'^{(e,i)}_{y'} \end{Bmatrix} = \begin{bmatrix} \dots & \mathbf{N}^{(e,i)} & \dots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{f}}'^{(e,i)}_{\mathrm{S}} \\ \vdots \end{Bmatrix}$$

Finite element discretization

Rotation matrix (global to local coordinates) of element (e):

- Given the nodal coordinates $\{\hat{\boldsymbol{x}}^{(e,i)}\}$ in the global system (x, y, z):

  1) Compute the surface vector: $\boldsymbol{S}^{(e)} = \left(\hat{\boldsymbol{x}}^{(e,3)} - \hat{\boldsymbol{x}}^{(e,1)}\right) \times \left(\hat{\boldsymbol{x}}^{(e,4)} - \hat{\boldsymbol{x}}^{(e,2)}\right)/2$

  2) Obtain normal: $\hat{\boldsymbol{k}}'^{(e)} = \boldsymbol{S}^{(e)} / \left\|\boldsymbol{S}^{(e)}\right\|$ ⟵ Area of the shell element

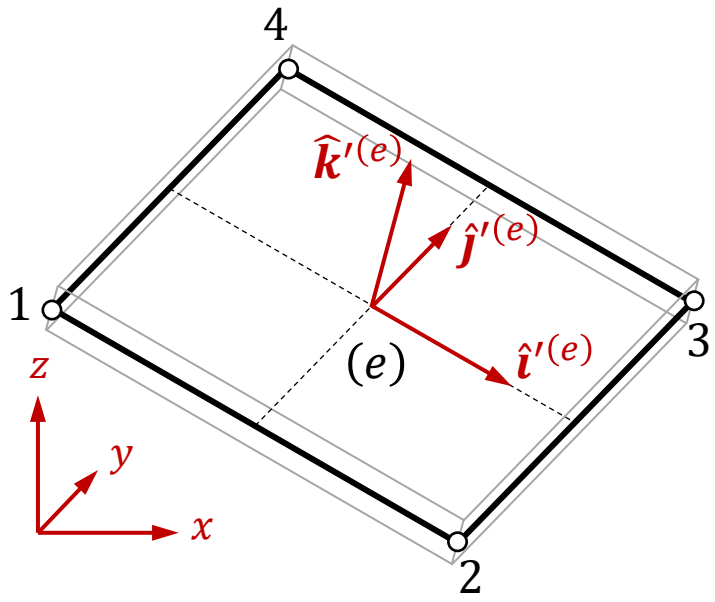
  3) Obtain middle segment: $\boldsymbol{d}^{(e)} = \left(\hat{\boldsymbol{x}}^{(e,2)} + \hat{\boldsymbol{x}}^{(e,3)} - \hat{\boldsymbol{x}}^{(e,4)} - \hat{\boldsymbol{x}}^{(e,1)}\right)/2$

  4) Obtain in-plane axes orientation: $\hat{\boldsymbol{\imath}}'^{(e)} = \boldsymbol{d}^{(e)} / \left\|\boldsymbol{d}^{(e)}\right\|$; $\hat{\boldsymbol{\jmath}}'^{(e)} = \hat{\boldsymbol{k}}'^{(e)} \times \hat{\boldsymbol{\imath}}'^{(e)}$

- The rotation matrix is then obtained as:

$$\left[\mathbf{R}^{(e)}\right] = \left[\hat{\boldsymbol{\imath}}'^{(e)} \quad \hat{\boldsymbol{\jmath}}'^{(e)} \quad \hat{\boldsymbol{k}}'^{(e)}\right]^{\mathrm{T}} \rightarrow \quad \{\hat{\boldsymbol{a}}'\} = \left[\mathbf{R}^{(e)}\right]\{\hat{\boldsymbol{a}}\}$$

$$\underbrace{\begin{Bmatrix} \hat{u}_{x'}^{(e,i)} \\ \hat{u}_{y'}^{(e,i)} \\ \hat{u}_{z'}^{(e,i)} \\ \hat{\theta}_{x'}^{(e,i)} \\ \hat{\theta}_{y'}^{(e,i)} \end{Bmatrix}}_{\{\hat{\boldsymbol{u}}'^{(e,i)}\}} = \underbrace{\begin{bmatrix} i_x'^{(e)} & i_y'^{(e)} & i_z'^{(e)} & 0 & 0 & 0 \\ j_x'^{(e)} & j_y'^{(e)} & j_z'^{(e)} & 0 & 0 & 0 \\ k_x'^{(e)} & k_y'^{(e)} & k_z'^{(e)} & 0 & 0 & 0 \\ 0 & 0 & 0 & i_x'^{(e)} & i_y'^{(e)} & i_z'^{(e)} \\ 0 & 0 & 0 & j_x'^{(e)} & j_y'^{(e)} & j_z'^{(e)} \end{bmatrix}}_{[\hat{\mathbf{R}}^{(e)}]} \underbrace{\begin{Bmatrix} \hat{u}_x^{(e,i)} \\ \hat{u}_y^{(e,i)} \\ \hat{u}_z^{(e,i)} \\ \hat{\theta}_x^{(e,i)} \\ \hat{\theta}_y^{(e,i)} \\ \hat{\theta}_z^{(e,i)} \end{Bmatrix}}_{\{\hat{\boldsymbol{u}}^{(e,i)}\}}$$



Finite element discretization

Discretized equation:

$$\sum_e \{\delta\hat{\boldsymbol{u}}^{(e)}\}^{\mathrm{T}} \left[[\mathbf{M}^{(e)}]\{\ddot{\hat{\boldsymbol{u}}}^{(e)}\} + \left(\left[\mathbf{K}_{\mathrm{m}}^{(e)}\right] + \left[\mathbf{K}_{\mathrm{b}}^{(e)}\right] + \left[\mathbf{K}_{\mathrm{s}}^{(e)}\right] \right) \{\hat{\boldsymbol{u}}^{(e)}\} - \{\hat{\boldsymbol{f}}^{(e)}\} \right] = 0$$

where:

$$\left[\mathbf{M}^{(e)}\right] = \left[\hat{\mathbf{R}}^{(e)}\right]^{\mathrm{T}} \int_{S^{(e)}} \left[\mathbf{N}^{(e)}\right]^{\mathrm{T}} \left[\bar{\boldsymbol{\rho}}'^{(e)}\right] \left[\mathbf{N}^{(e)}\right] \mathrm{d}S \left[\hat{\mathbf{R}}^{(e)}\right]$$

$$\left[\mathbf{K}_{\mathrm{m}}^{(e)}\right] = \left[\hat{\mathbf{R}}^{(e)}\right]^{\mathrm{T}} \int_{S^{(e)}} \left[\mathbf{B}_{\mathrm{m}}'^{(e)}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}_{\mathrm{m}}'^{(e)}\right] \left[\mathbf{B}_{\mathrm{m}}'^{(e)}\right] \mathrm{d}S \left[\hat{\mathbf{R}}^{(e)}\right]$$

$$\left[\mathbf{K}_{\mathrm{b}}^{(e)}\right] = \left[\hat{\mathbf{R}}^{(e)}\right]^{\mathrm{T}} \int_{S^{(e)}} \left[\mathbf{B}_{\mathrm{b}}'^{(e)}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}_{\mathrm{b}}'^{(e)}\right] \left[\mathbf{B}_{\mathrm{b}}'^{(e)}\right] \mathrm{d}S \left[\hat{\mathbf{R}}^{(e)}\right]$$

$$\left[\mathbf{K}_{\mathrm{s}}^{(e)}\right] = \left[\hat{\mathbf{R}}^{(e)}\right]^{\mathrm{T}} \int_{S^{(e)}} \left[\mathbf{B}_{\mathrm{s}}'^{(e)}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}_{\mathrm{s}}'^{(e)}\right] \left[\mathbf{B}_{\mathrm{s}}'^{(e)}\right] \mathrm{d}S \left[\hat{\mathbf{R}}^{(e)}\right]$$

External surface loads (N/m²)

$$\{\hat{\boldsymbol{p}}^{(e)}\} = \{\hat{\boldsymbol{q}}^{(e)}\} + \left\{\hat{\boldsymbol{t}}_{\mathrm{S}}^{(e)}\right\}$$

$$\{\hat{\boldsymbol{f}}^{(e)}\} = [\mathbf{M}^{(e)}]\{\hat{\boldsymbol{b}}^{(e)}\} + \left[\hat{\mathbf{R}}^{(e)}\right]^{\mathrm{T}} \left(\int_{S^{(e)}} \left[\mathbf{N}^{(e)}\right]^{\mathrm{T}} \left[\mathbf{N}^{(e)}\right] \mathrm{d}S \left[\hat{\mathbf{R}}^{(e)}\right] \{\hat{\boldsymbol{p}}^{(e)}\} + \int_{\ell^{(e)}} \left[\mathbf{N}^{(e)}\right]^{\mathrm{T}} \left[\mathbf{N}^{(e)}\right] \mathrm{d}S \left[\hat{\mathbf{R}}^{(e)}\right] \left\{\hat{\boldsymbol{t}}_{\ell}^{(e)}\right\} \right)$$

Global matrix system:

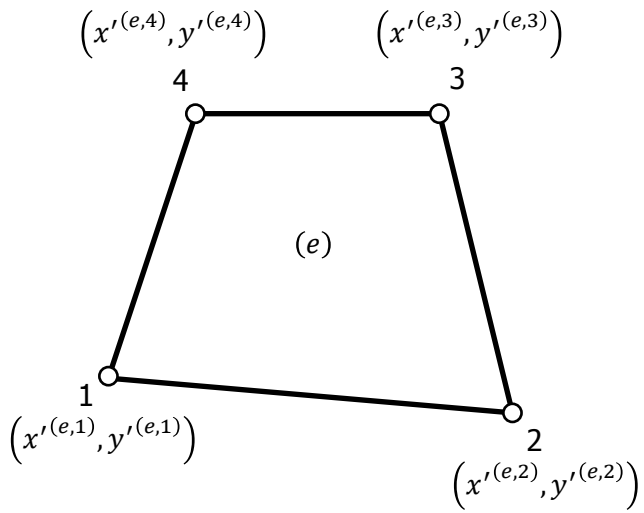
$$[\mathbf{M}]\{\ddot{\hat{\boldsymbol{u}}}\} + ([\mathbf{K}_{\mathrm{a}}] + [\mathbf{K}_{\mathrm{b}}] + [\mathbf{K}_{\mathrm{s}}])\{\hat{\boldsymbol{u}}\} = \{\hat{\boldsymbol{f}}\}$$

Quadrilateral elements integration

Shape functions for quadrilateral elements:

$$N^{(e,i)}(\xi,\eta) = \frac{1}{4}(1+a_i\xi)(1+b_i\eta)$$

i	1	2	3	4
a_i	-1	1	1	-1
b_i	-1	-1	1	1



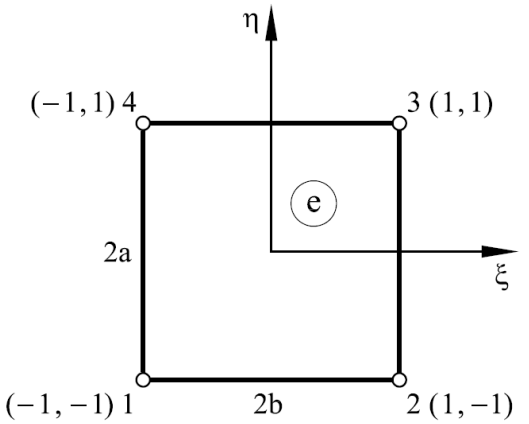
Shape functions *natural* derivatives:

$$N^{(e,i)}_{,\xi}(\xi,\eta) = \frac{1}{4}a_i(1+b_i\eta)$$

$$N^{(e,i)}_{,\eta}(\xi,\eta) = \frac{1}{4}b_i(1+a_i\xi)$$

Jacobian matrix:

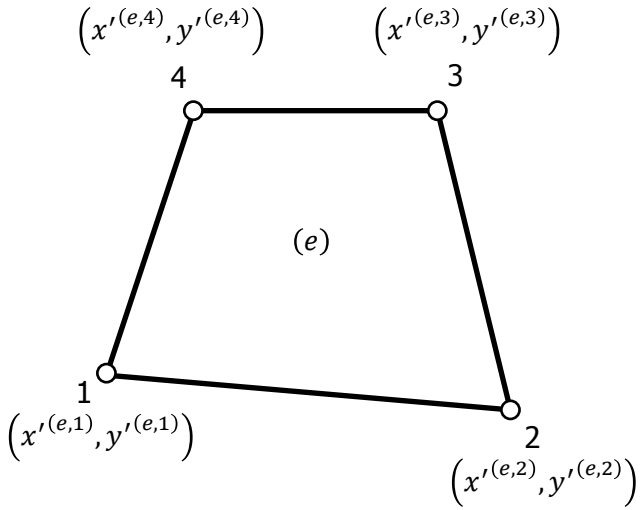
$$\left[\mathcal{J}'^{(e)}\right] = \begin{bmatrix} \frac{\partial x'}{\partial \xi} & \frac{\partial y'}{\partial \xi} \\ \frac{\partial x'}{\partial \eta} & \frac{\partial y'}{\partial \eta} \end{bmatrix} = \sum_i \begin{Bmatrix} N^{(e,i)}_{,\xi} \\ N^{(e,i)}_{,\eta} \end{Bmatrix} \overbrace{\left\{\hat{\boldsymbol{x}}^{(e,i)}\right\}^{\mathrm{T}}}^{\left\{\hat{\boldsymbol{x}}'^{(e,i)}\right\}^{\mathrm{T}}} \left[\hat{\boldsymbol{\imath}}'^{(e)} \quad \hat{\boldsymbol{\jmath}}'^{(e)}\right]$$



Quadrilateral elements integration

Shape functions *local* derivatives:

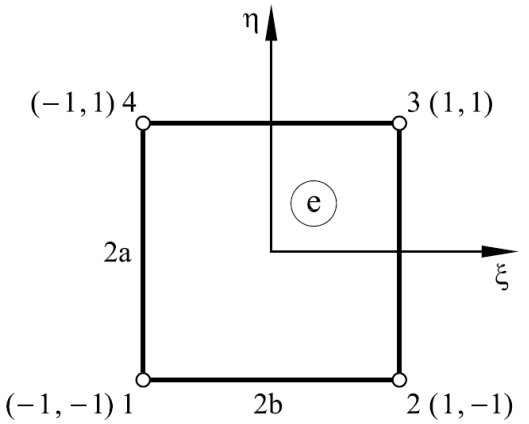
$$\begin{Bmatrix} N^{(e,i)}_{,x'} \\ N^{(e,i)}_{,y'} \end{Bmatrix} = \left[\mathcal{J}'^{(e)}\right]^{-1} \begin{Bmatrix} N^{(e,i)}_{,\xi} \\ N^{(e,i)}_{,\eta} \end{Bmatrix}$$



With $N^{(e,i)}, N^{(e,i)}_{,x'}$ and $N^{(e,i)}_{,y'}$, one can compute $\left[\mathbf{N}^{(e)}\right]$, $\left[\mathbf{B}'^{(e)}_{\mathrm{m}}\right]$, $\left[\mathbf{B}'^{(e)}_{\mathrm{b}}\right]$, $\left[\mathbf{B}'^{(e)}_{\mathrm{s}}\right]$ and $\left[\mathbf{B}'^{(e)}_{\mathrm{t}}\right]$, in terms of the *natural coordinates* (ξ, η).

To evaluate the required integrals, a **Gauss quadrature** may be performed. For N_G gauss points, at positions (ξ_k, η_k) with weights w_k $(k = 1 \ldots n_G)$. For example:

$$\int_{S^{(e)}} \left[\mathbf{B}'^{(e)}_{\mathrm{m}}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}'^{(e)}_{\mathrm{m}}\right] \left[\mathbf{B}'^{(e)}_{\mathrm{m}}\right] \mathrm{d}S = \sum_{k=1}^{N_G} w_k \left|\mathcal{J}'^{(e)}\right| \left[\mathbf{B}'^{(e)}_{\mathrm{m}}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}'^{(e)}_{\mathrm{m}}\right] \left[\mathbf{B}'^{(e)}_{\mathrm{m}}\right]$$

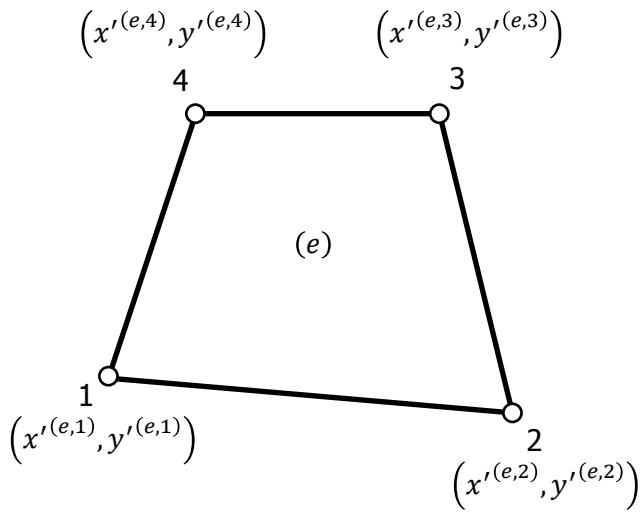


where $\left|\mathcal{J}'^{(e)}\right|$ is the determinant of the Jacobian matrix.

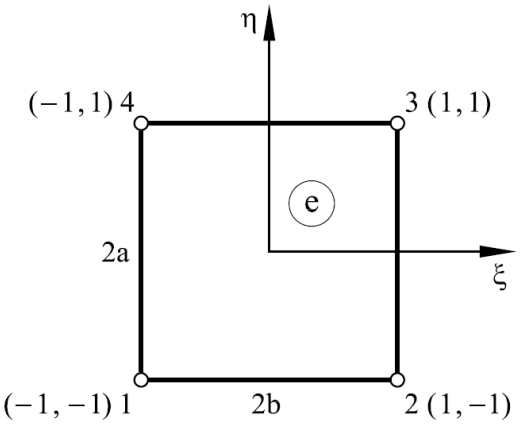
Notice that, in general, $\left|\mathcal{J}'^{(e)}\right|$ and $\left[\mathbf{B}'^{(e)}_{\mathrm{m}}\right]$ will be a function of (ξ_k, η_k).

Quadrilateral elements integration

Remark. The approach considered is general for thick and thin plates/shells modelled with flat 4-noded quadrilateral elements. However, it suffers from the ***shear-locking*** issue for some special cases. To improve the performance of the quadrilateral shell element, it is recommended to sub-integrate (with just 1 Gauss point instead of 4) the shear components of the stiffness matrices.

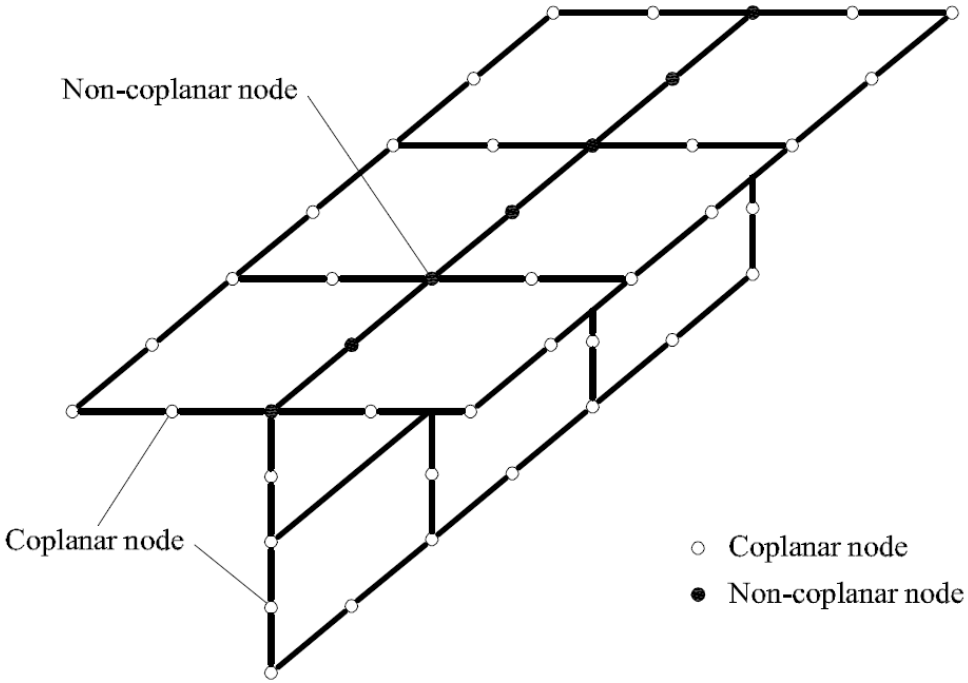


N_G	k	w_k	ξ_k	η_k
1	1	4	0	0
4	1	1	$-1/\sqrt{3}$	$-1/\sqrt{3}$
	2	1	$+1/\sqrt{3}$	$-1/\sqrt{3}$
	3	1	$+1/\sqrt{3}$	$+1/\sqrt{3}$
	4	1	$-1/\sqrt{3}$	$+1/\sqrt{3}$



Coplanar nodes

Another issue appearing in 3D configurations is the problem with so-called coplanar nodes. The problem arises because the model considers only 5 degrees of freedom in the local reference frame, while in the global 3D setting, we will have 6. This means that for each coplanar node, we will end up having an additional equation in the global system that is either 0, or linearly dependent with another, making the global stiffness matrix to be singular.



Coplanar nodes

To deal with this issue, once the coplanar nodes and elements containing them have been identified, an additional stiffness component associated to the $\theta_{z'}^{(e,i)}$ degree of freedom is added to the global stiffness matrix. Essentially, we are adding an equation:

$$K_{\theta_{z'}}\theta_{z'}^{(e,i)} = 0$$

$$\theta_{z'}^{(e,i)} = \begin{Bmatrix} k_x'^{(e)} & k_y'^{(e)} & k_z'^{(e)} \end{Bmatrix} \begin{Bmatrix} \hat{\theta}_x^{(e,i)} \\ \hat{\theta}_y^{(e,i)} \\ \hat{\theta}_z^{(e,i)} \end{Bmatrix}$$

So, the element stiffness matrix components corresponding to the element's coplanar nodes is now:

$$\left[\overline{\mathbf{K}}^{(e,i)}\right] = \left[\mathbf{K}^{(e,i)}\right] + \left[\mathbf{K}_{\mathrm{r}}^{(e,i)}\right]; \quad \left[\mathbf{K}_{\mathrm{r}}^{(e,i)}\right] = K_{\theta_{z'}} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & k_x'^{(e)}k_x'^{(e)} & k_x'^{(e)}k_y'^{(e)} & k_x'^{(e)}k_z'^{(e)} \\ 0 & 0 & 0 & k_y'^{(e)}k_x'^{(e)} & k_y'^{(e)}k_y'^{(e)} & k_y'^{(e)}k_z'^{(e)} \\ 0 & 0 & 0 & k_z'^{(e)}k_x'^{(e)} & k_z'^{(e)}k_y'^{(e)} & k_z'^{(e)}k_z'^{(e)} \end{bmatrix}$$

Coplanar nodes

To deal with this issue, once the coplanar nodes and elements containing them have been identified, an additional stiffness component associated to the $\theta_{z'}^{(e,i)}$ degree of freedom is added to the global stiffness matrix. Essentially, we are adding an equation:

$$K_{\theta_{z'}}\theta_{z'}^{(e,i)} = 0$$

$$\left\{\begin{matrix}\hat{\theta}_x^{(e,i)}\\ \hat{\theta}_y^{(e,i)}\\ \hat{\theta}_z^{(e,i)}\end{matrix}\right\}$$

Since the local equation added is uncoupled from the rest, the results should not be affected by the value of $K_{\theta_{z'}}$. To help conditioning the global stiffness matrix, typically:

$$K_{\theta_{z'}} = E^{(e)}h^{(e)}S^{(e)}$$

onding to the element's coplanar

$$\left[\bar{\mathbf{K}}^{(e,i)}\right] = \left[\mathbf{K}^{(e,i)}\right] + \left[\mathbf{K}_{\mathrm{r}}^{(e,i)}\right];\quad \left[\mathbf{K}_{\mathrm{r}}^{(e,i)}\right] = K_{\theta_{z'}}\begin{bmatrix} & & & 0 & 0 & 0 & 0\\ & & & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & k_x'^{(e)}k_x'^{(e)} & k_x'^{(e)}k_y'^{(e)} & k_x'^{(e)}k_z'^{(e)} & \\ 0 & 0 & 0 & k_y'^{(e)}k_x'^{(e)} & k_y'^{(e)}k_y'^{(e)} & k_y'^{(e)}k_z'^{(e)} & \\ 0 & 0 & 0 & k_z'^{(e)}k_x'^{(e)} & k_z'^{(e)}k_y'^{(e)} & k_z'^{(e)}k_z'^{(e)} & \end{bmatrix}$$